Metal–Semiconductor Interfaces

ELE 717 Solid State Electronics · Week 11 · Group presentation notes

Subtopics: (1) Schottky barrier diode physics versus ohmic contact mechanisms; (2) energy band diagrams under equilibrium and applied bias.

Learning objectives. After this topic you should be able to:

1. Define work function, electron affinity and Fermi level, and relate them for a doped semiconductor.
2. Draw the energy band diagram of a metal and an n-type semiconductor before and after contact.
3. Explain how a Schottky barrier forms, and calculate the barrier height, built-in potential and depletion width.
4. Draw and explain the band diagrams under forward and reverse bias, and use the thermionic-emission equation.
5. Explain the three ways of making an ohmic contact, and why heavy doping works.
6. Compare Schottky and ohmic contacts, and answer examination question 49.

Units. All quantities are in SI units. Energies are quoted in electron-volts (eV), which SI accepts for use alongside its units ($1\ \text{eV} = 1.6 \times 10^{-19}$ J). A work function Φ is a potential in volts; $q\Phi$ is the corresponding energy, and its value in eV equals the value of Φ in volts.

1. Introduction

Every semiconductor device has to be connected to the outside world with metal. Where a metal touches a semiconductor, a **metal–semiconductor interface** (or **contact**) is formed. When the two materials touch, electrons move from one to the other until a single Fermi level runs through both. This charge transfer can leave an energy barrier at the interface, and that barrier decides how current flows. There are two kinds of contact:

- A **Schottky contact** has a barrier to majority carriers, so it is **rectifying**: current flows easily in one direction (forward bias) and very little in the other (reverse bias). A diode made this way is a **Schottky barrier diode (SBD)**. The rectifying metal–semiconductor contact was explained by Walter Schottky and Nevill Mott in 1938–39.
- An **ohmic contact** has no barrier that limits the current, so it conducts equally in both directions with a linear current–voltage (I–V) graph. Every transistor terminal, solar-cell finger and integrated-circuit connection needs one.

A practical Schottky diode contains both: a rectifying contact on top and an ohmic contact underneath (Figure 1).

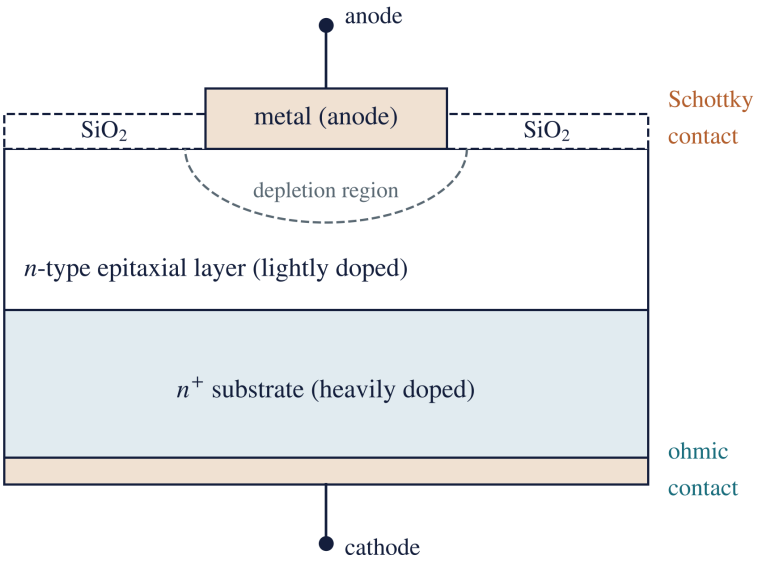


Figure 1. A Schottky barrier diode in cross-section. The metal anode on a lightly doped n-type epitaxial layer forms the rectifying Schottky contact, with a depletion region beneath it. The heavily doped n^+ substrate and the metal underneath form the ohmic cathode contact. The oxide ring insulates the edges of the metal.

2. Key terms and definitions

Metal. A material with a very large density of free electrons (10^{28}–10^{29} m^{-3}; copper has 8.5×10^{28} m^{-3}), so it conducts very well. Its electron states are filled up to the Fermi level, with empty states immediately above. Examples: aluminium, gold, platinum, titanium, tungsten.

Semiconductor. A material whose conductivity lies between that of a metal and an insulator and can be controlled by doping, temperature, light and electric field. Examples: silicon (Si), germanium (Ge), gallium arsenide (GaAs).

Doping; n-type and p-type. Adding a small, controlled amount of impurity. Donor atoms (concentration N_D, e.g. phosphorus in Si) give free electrons and make the material **n-type**; acceptor atoms (concentration N_A, e.g. boron) create holes and make it **p-type**. The carrier in the majority (electrons in n-type, holes in p-type) is the **majority carrier**.

Band edges and band gap. E_c is the bottom of the conduction band (the lowest energy of a free electron); E_v is the top of the valence band. No electron states exist between them. The band gap is

$$E_g = E_c - E_v$$

where E_g is the band-gap energy (1.12 eV for Si at 300 K), E_c the conduction-band edge and E_v the valence-band edge.

Fermi level (E_F). The energy at which a state has a 50 % probability of being occupied by an electron. In n-type material it lies close to E_c, in p-type close to E_v. **At thermal equilibrium the Fermi level is flat: it has the same value everywhere in materials that are in contact.** For n-type silicon,

$$E_c - E_F = kT \ln\left(\frac{N_c}{N_D}\right)$$

where $kT = 0.0259$ eV at 300 K, $N_c = 2.8 \times 10^{25}$ m^{-3} is the effective density of states in the conduction band of Si, and N_D is the donor concentration in m^{-3}.

Vacuum level (E_{vac}). The energy of an electron at rest just outside the material. It is the common reference from which the other energies below are measured.

Work function (Φ). The minimum energy needed to take an electron from the Fermi level to the vacuum level:

$$q\Phi_m = E_{\text{vac}} - E_{F,m}, \qquad q\Phi_s = E_{\text{vac}} - E_{F,s}$$

where Φ_m and Φ_s are the work functions of the metal and of the semiconductor, $E_{F,m}$ and $E_{F,s}$ their Fermi levels, and $q = 1.6 \times 10^{-19}$ C the electronic charge. A metal with a large work function holds its electrons tightly.

Electron affinity (χ). The energy from the bottom of the conduction band to the vacuum level:

$$q\chi = E_{\text{vac}} - E_c$$

Unlike Φ_s, the electron affinity does not depend on doping; it is a fixed property of the material (Si 4.05 eV, GaAs 4.07 eV, Ge 4.0 eV). The two are related by

$$q\Phi_s = q\chi + (E_c - E_F)$$

so the semiconductor work function **does** change with doping, because doping moves E_F.

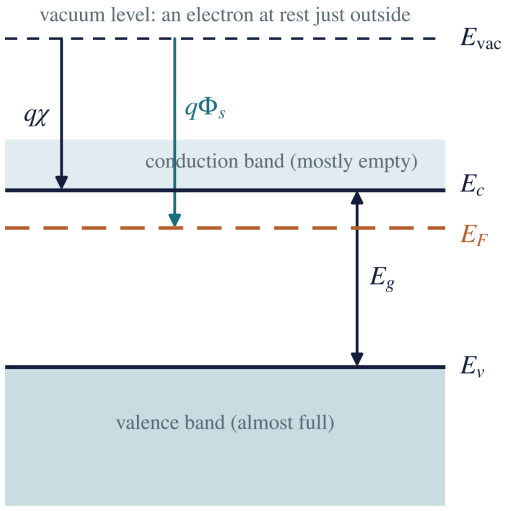


Figure 2. Energy levels of an isolated n-type semiconductor. The conduction band starts at E_c and is mostly empty; the valence band below E_v is almost full. The electron affinity $q\chi$ is measured from E_c to the vacuum level, the work function $q\Phi_s$ from E_F.

Table 1. Work functions of common contact metals and electron affinities of semiconductors.

Metal	Al	Ti	Cr	W	Au	Ni	Pt
Φ_m (eV)	4.28	4.33	4.5	4.55	5.10	5.15	5.65

Semiconductor	Si	Ge	GaAs
χ (eV)	4.05	4.0	4.07
E_g (eV)	1.12	0.66	1.42

Depletion region. A layer next to the interface emptied of free carriers, leaving only the fixed, charged dopant ions (positive donor ions in n-type). **Accumulation layer:** the opposite, a thin layer with extra majority carriers.

Built-in potential (V_{bi}). The voltage across the depletion region at equilibrium.

Thermionic emission. Carriers crossing **over** a barrier using their thermal energy. **Tunnelling (field emission):** carriers passing **through** a barrier thin enough (a few nanometres), a quantum-mechanical effect.

3. Energy band diagram before contact

Before the materials touch, each has its own Fermi level. Lining them up against the common vacuum level (Figure 3) shows which way electrons will move on contact. For a metal on **n-type** silicon with $\boldsymbol{\Phi_m > \Phi_s}$, the semiconductor's Fermi level lies higher than the metal's by $q(\boldsymbol{\Phi_m - \Phi_s})$. Electrons always move from the higher Fermi level to the lower one, so on contact they will flow **from the semiconductor into the metal**.

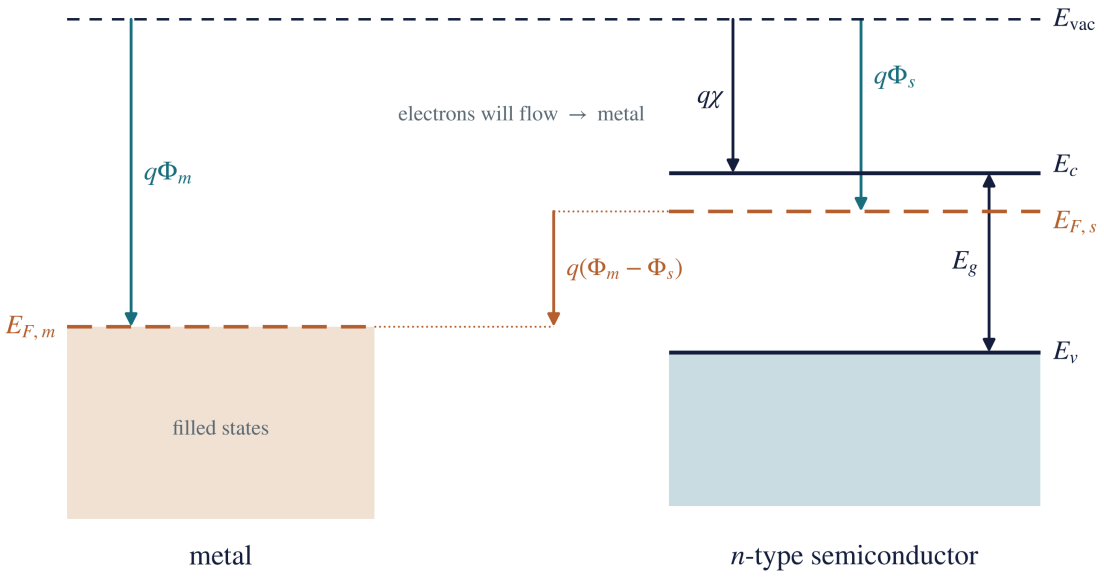


Figure 3. A metal and an n-type semiconductor before contact, measured from the common vacuum level. With $\boldsymbol{\Phi_m > \Phi_s}$, the semiconductor's Fermi level lies higher by $q(\boldsymbol{\Phi_m - \Phi_s})$.

Which contact forms depends on the two work functions (ideal case):

Semiconductor	$\boldsymbol{\Phi_m > \Phi_s}$	$\boldsymbol{\Phi_m < \Phi_s}$
n-type	Schottky (rectifying): depletion, bands bend up	Ohmic: accumulation, bands bend down
p-type	Ohmic: hole accumulation	Schottky (rectifying): bands bend down, barrier to holes

4. How a Schottky barrier forms

Consider an ideal metal on n-type silicon with $\boldsymbol{\Phi_m > \Phi_s}$, and no interface states (the **Schottky–Mott** model). When contact is made:

1. **Electrons flow from the semiconductor into the metal**, because the semiconductor's Fermi level is higher.

2. **Positive donor ions are left behind.** Near the interface the semiconductor loses its free electrons, uncovering the fixed ionised donors N_D^+.
3. **A depletion region of width W forms**, holding positive charge; an equal negative charge sits as a thin sheet of extra electrons on the metal surface.
4. **An electric field appears** between the positive ions and the negative sheet, pointing from the semiconductor towards the metal.
5. **The bands bend upward** towards the interface, because the field raises the electron energy near the metal.
6. **The flow stops when the Fermi levels line up.** At equilibrium there is a single, flat Fermi level: $E_{F,m} = E_{F,s}$. Electrons still cross in both directions, but the two flows are equal, so the net current is zero.

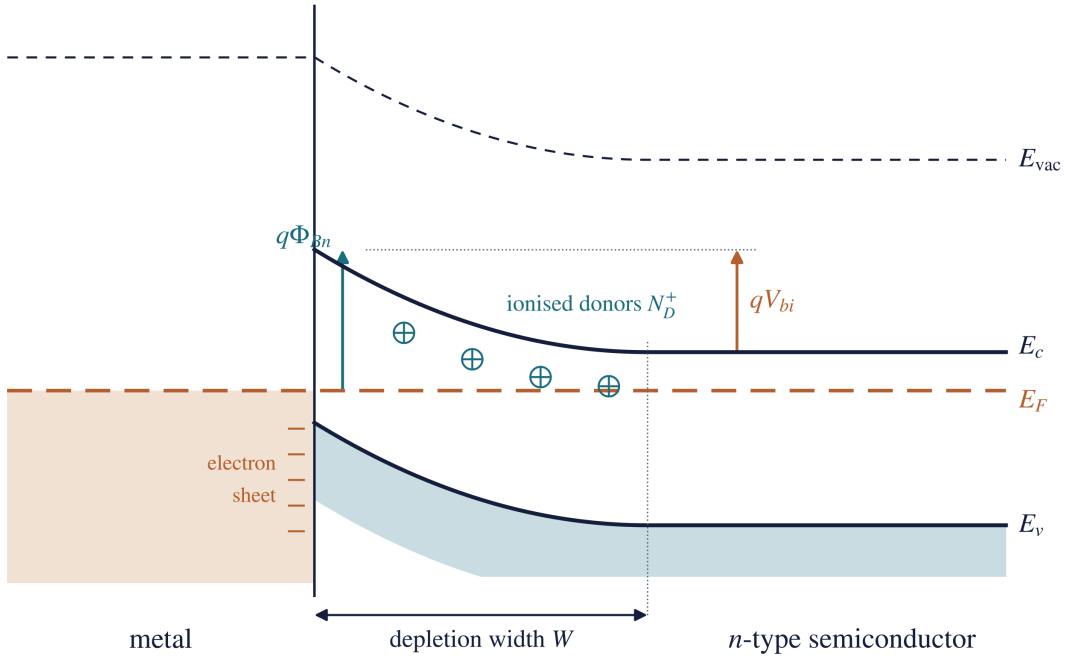


Figure 4. A Schottky contact at equilibrium. The Fermi level is flat; the semiconductor bands bend upward over the depletion width W. An electron in the metal faces the barrier $q\Phi_{Bn}$; an electron in the semiconductor faces the smaller built-in barrier qV_{bi}. Ionised donors fill the depletion region and a sheet of electrons sits on the metal surface.

4.1 Barrier height and built-in potential

There are now two barriers, depending on which side the electron starts from (Figure 4).

Barrier height (seen from the metal). Because the vacuum level is continuous at the interface, the conduction-band edge at the interface sits at $q\chi$ below the vacuum level, while the metal Fermi level sits at $q\Phi_m$ below it:

$$q\Phi_{Bn} = q(\Phi_m - \chi)$$

where Φ_{Bn} is the Schottky barrier height for electrons (n-type), Φ_m the metal work function and χ the electron affinity. This is the **Schottky–Mott rule**. The barrier height is set by the interface and **does not change with applied bias**.

Built-in potential (seen from the semiconductor). The total band bending equals the original difference in Fermi levels:

$$qV_{bi} = q(\Phi_m - \Phi_s) = q\Phi_{Bn} - (E_c - E_F)$$

p-type semiconductor. A rectifying contact needs $\Phi_m < \Phi_s$, and the barrier is for holes:

$$q\Phi_{Bp} = E_g - q(\Phi_m - \chi)$$

so that, for the same metal, $q\Phi_{Bn} + q\Phi_{Bp} = E_g$.

4.2 The depletion region

With the depletion approximation (no free carriers inside $0 < x < W$, uniform doping), the charge density is $\rho = qN_D$ in the depletion region. Integrating Poisson's equation once gives a field that is largest at the interface and falls linearly to zero at $x = W$; integrating again gives the parabolic band bending (Figure 5). The results are

$$W = \sqrt{\frac{2\varepsilon_s (V_{bi} - V)}{qN_D}}, \qquad \mathscr{E}_{\max} = \frac{qN_D W}{\varepsilon_s}$$

where W is the depletion width (m), $\varepsilon_s = 11.7\varepsilon_0 = 1.04 \times 10^{-10}$ F m^{-1} the permittivity of silicon, V the applied voltage (positive for forward bias), N_D the donor concentration (m^{-3}) and $\mathscr{E}_{\max}$ the peak electric field at the interface (V m^{-1}). The depletion region acts like a parallel-plate capacitor of capacitance per unit area $C = \varepsilon_s / W$, which is the basis of C–V measurement (Week 12).

Two consequences matter later: $W \propto (V_{bi} - V)^{1/2}$, so reverse bias widens the region; and $W \propto 1/\sqrt{N_D}$, so heavy doping makes it very thin.

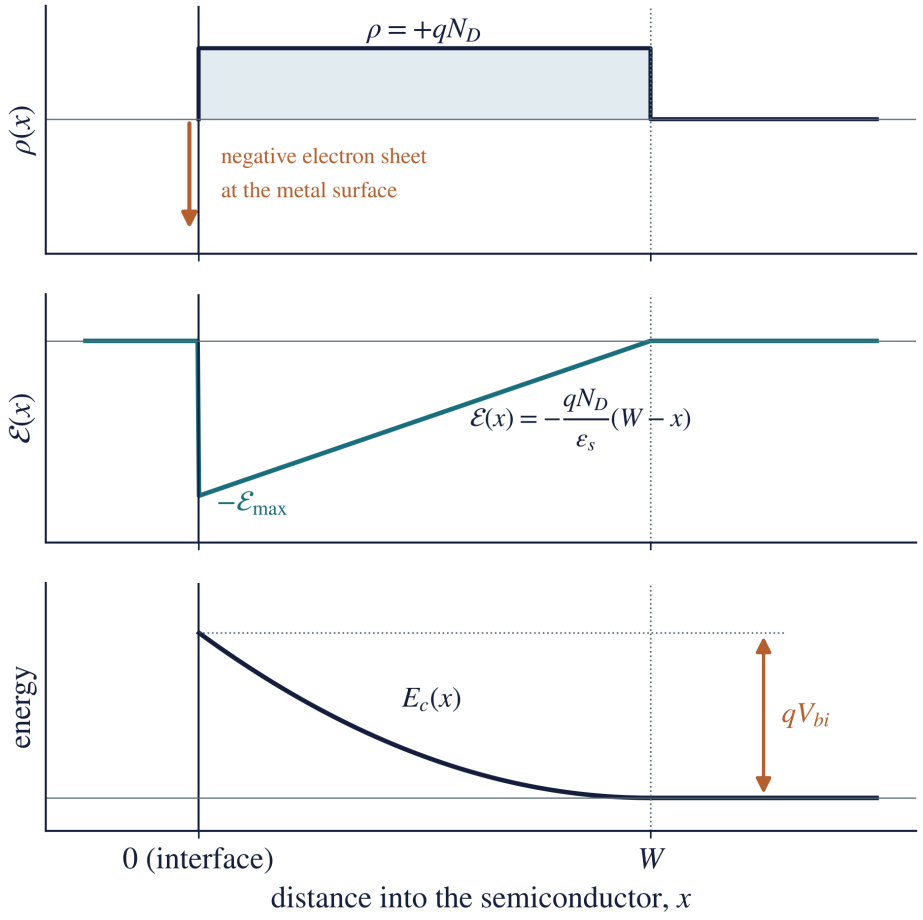


Figure 5. The depletion region of a Schottky contact: charge density (positive donors, with the balancing electron sheet in the metal), electric field (largest at the interface) and conduction-band edge (bent up by qV_{bi}).

4.3 Real interfaces: Fermi-level pinning and image-force lowering

Fermi-level pinning. Measured barrier heights depend much less on the metal than the Schottky–Mott rule predicts (Figure 6). The reason is **interface states**: dangling bonds and defects at the surface create electron

states in the band gap. They hold, or "pin", the Fermi level near a fixed energy at the surface, whatever metal is used. On n-Si most barriers lie between 0.5 and 0.9 eV; on n-GaAs almost every metal gives about 0.8 eV. **In calculations, use a measured barrier height when one is given.**

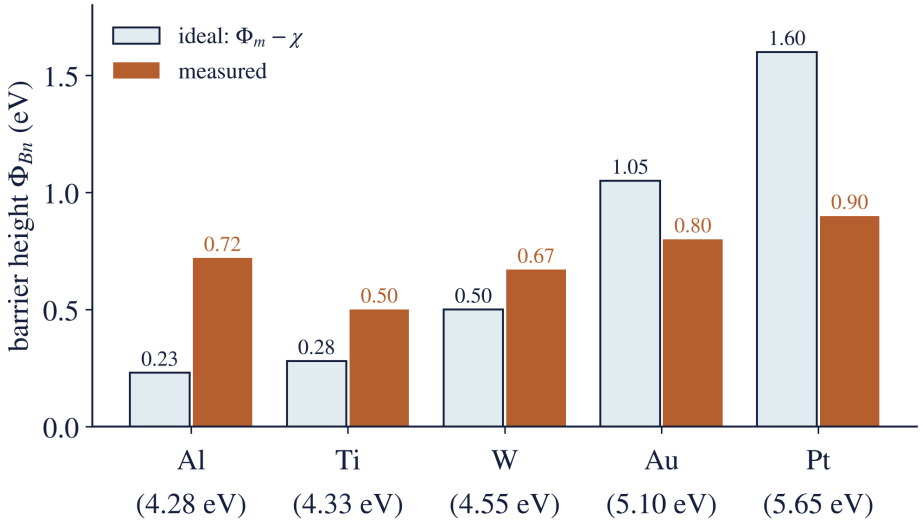


Figure 6. Ideal ($\Phi_m - \chi$) and typical measured barrier heights for metals on n-type silicon. The measured values vary much less than the ideal ones: the Fermi level is pinned by interface states. Measured values are typical literature values (Sze and Ng).

Image-force lowering. An electron near the metal is attracted by the positive "image" charge it induces in the metal. This rounds off the top of the barrier and lowers it slightly by

$$\Delta\Phi = \sqrt{\frac{q\,\mathscr{E}_{\max}}{4\pi\varepsilon_s}}$$

For the gold contact of Example 2 below ($\mathscr{E}_{\max} = 5.1 \times 10^6$ V m^{-1}), $\Delta\Phi \approx 0.025$ V: small, but it grows with reverse bias and is one reason the reverse current of a Schottky diode does not fully saturate.

5. Energy band diagrams under applied bias

The applied voltage is defined as $V = V_m - V_s$, the potential of the metal terminal minus that of the semiconductor terminal: **forward bias** means $V > 0$ (metal positive, for n-type) and **reverse bias** means $V < 0$, written $V = -V_R$. When a voltage is applied, the two sides are no longer in equilibrium, so there is no single flat Fermi level: the semiconductor's Fermi level moves by qV relative to the metal's. **The key rule: the barrier seen from the metal, $q\Phi_{Bn}$, stays fixed; the applied voltage appears across the depletion region and changes only the barrier seen from the semiconductor** (Figure 7).

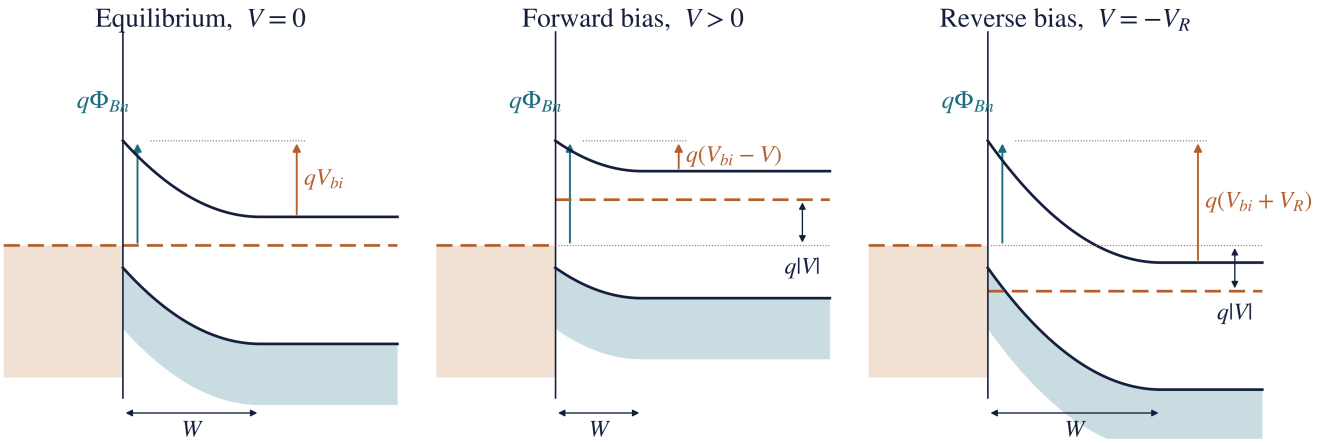


Figure 7. The Schottky contact at equilibrium, under forward bias and under reverse bias. The barrier from the metal, $q\Phi_{Bn}$, does not change. Forward bias lowers the semiconductor-side barrier to $q(V_{bi} - V)$ and narrows W; reverse bias raises it to $q(V_{bi} + V_R)$ and widens W.

	Equilibrium, $V = 0$	**Forward bias, $V > 0$ (metal positive)**	**Reverse bias, $V = -V_R$ (metal negative)**
Fermi levels	one flat E_F	semiconductor E_F raised by qV	semiconductor E_F lowered by qV_R
Band bending	qV_{bi}	$q(V_{bi} - V)$, smaller	$q(V_{bi} + V_R)$, larger
Depletion width	W_0	narrower	wider
Electrons semiconductor → metal	balance the reverse flow	increase by $\exp(qV/kT)$	almost stopped
Electrons metal → semiconductor	small, over $q\Phi_{Bn}$	unchanged	unchanged: this is the saturation current
Net current	zero	large, rises exponentially	small, about $-I_s$

In the ideal model the reverse current approaches $-I_s$ once the reverse voltage is a few times kT/q (about 0.1 V). At large reverse voltage the field becomes strong enough for **breakdown**. Schottky diodes break down at lower voltages than comparable p–n diodes.

6. Current transport: thermionic emission

In a Schottky diode the current is carried by **majority carriers** (electrons in n-type) crossing **over** the barrier. Only electrons with enough thermal energy get over, so the current depends exponentially on the barrier height and on temperature:

$$J = J_s\left[\exp\left(\frac{qV}{\eta kT}\right) - 1\right], \qquad J_s = A^{**}T^2\exp\left(-\frac{q\Phi_{Bn}}{kT}\right)$$

where

- J is the current density ($\mathrm{A\,m^{-2}}$); the current is $I = JA$ for a contact of area A, so $I = I_s\left[\exp(qV/\eta kT) - 1\right]$ with $I_s = J_s A$;
- J_s is the saturation current density ($\mathrm{A\,m^{-2}}$);
- V is the applied voltage, positive for forward bias (V);
- η is the ideality factor: 1 for ideal thermionic emission, 1.0–1.1 for a good diode (some books write it as n);

- $k = 1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ is the Boltzmann constant ($kT/q = 0.0259$ V at 300 K) and T the absolute temperature (K);
- A^{**} is the effective Richardson constant, about $1.1 \times 10^{6}\ \mathrm{A\,m^{-2}\,K^{-2}}$ for n-type Si.

The I–V curve has the same shape as a p–n diode's, but J_s is typically 10^3–10^8 times larger, so the Schottky diode turns on at a lower voltage (about 0.2–0.4 V on Si, against 0.6–0.7 V) and has a larger reverse leakage current (Figure 8). Other transport routes add to thermionic emission in real diodes: tunnelling through the top of the barrier at heavy doping (thermionic-field emission), recombination in the depletion region, and the voltage drop across the series resistance.

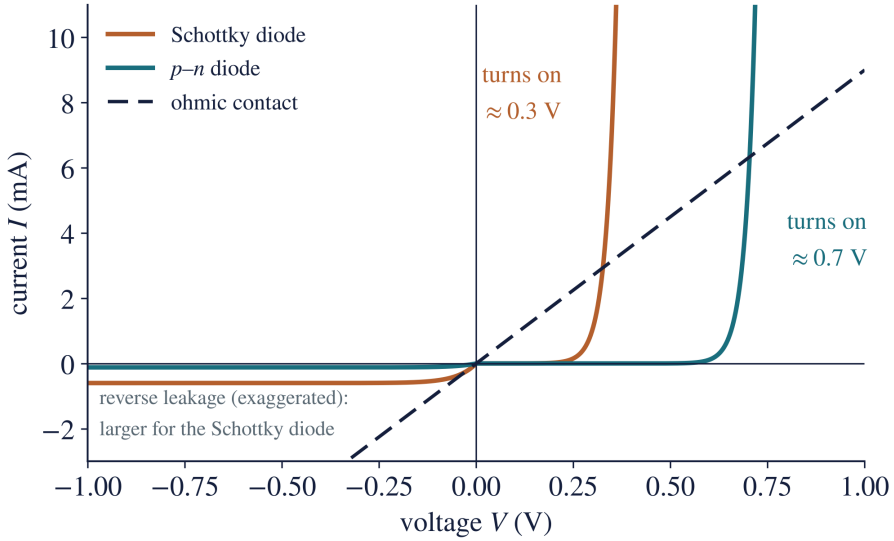


Figure 8. Current–voltage curves of a Schottky diode, a silicon p–n diode and an ohmic contact. The Schottky diode turns on at a lower voltage but leaks more in reverse (reverse currents exaggerated for visibility).

6.1 Schottky diode versus p–n diode

	Schottky diode	p–n junction diode
Current carried by	majority carriers only	minority carriers injected and diffusing
Transport mechanism	thermionic emission over the barrier	diffusion
Stored minority charge	practically none	yes, $Q \approx I\tau$
Switching speed	very fast, no reverse-recovery delay	slower (recovery time about τ)
Turn-on voltage (Si)	0.2–0.4 V	0.6–0.7 V
Reverse leakage	higher	lower
Breakdown voltage	lower	higher

Applications of Schottky contacts: fast, low-loss rectifiers in switching power supplies; RF and microwave detectors and mixers; clamping diodes that stop transistors saturating (Schottky TTL); gates of MESFETs and HEMTs; photodetectors and some solar-cell structures.

7. Ohmic contacts

An **ohmic contact** lets current flow equally easily in both directions, with a linear I–V characteristic and a resistance negligible compared with the device:

$$V = IR_c, \qquad R_c = \frac{\rho_c}{A}$$

where V is the voltage across the contact (V), I the current (A), R_c the contact resistance (Ω), A the contact area (m^2), and ρ_c the **specific contact resistance** ($\Omega\,\mathrm{m}^2$), which allows contacts of different sizes to be compared. Good contacts reach $\rho_c \approx 10^{-10}$–$10^{-12}\ \Omega\,\mathrm{m}^2$.

An ohmic contact is not necessarily a contact with no barrier. It may have a barrier that is so low, or so thin, that it does not restrict the current. There are three ways to make one (Figure 9).

Method 1: a low work-function metal (accumulation). For n-type, choose $\Phi_m < \Phi_s$ (for p-type, $\Phi_m > \Phi_s$). On contact, electrons flow from the metal **into** the semiconductor and pile up in an accumulation layer; the bands bend **downward** and there is no barrier to electrons in either direction. In practice, Fermi-level pinning makes suitable metals rare for Si and GaAs, so this method alone is unreliable.

Method 2: heavy doping, so that electrons tunnel (the practical method). Dope the semiconductor very heavily just under the contact (n^+ or p^+, above about $10^{25}\ \mathrm{m}^{-3}$). The barrier height does not change, but since

$$W \propto \frac{1}{\sqrt{N_D}}$$

the barrier becomes only a few nanometres thick. Electrons then **tunnel** straight through it in both directions (field emission). The specific contact resistance falls steeply, roughly exponentially, as the doping rises.

Method 3: contact engineering. Clean the surface (remove the native oxide), deposit the metal and anneal it so that it reacts with the semiconductor to form a stable compound with a low barrier: a **silicide** ($TiSi_2$, $CoSi_2$, NiSi) on silicon, or an alloyed Au–Ge–Ni contact on n-GaAs. This is combined with Method 2 in practice.

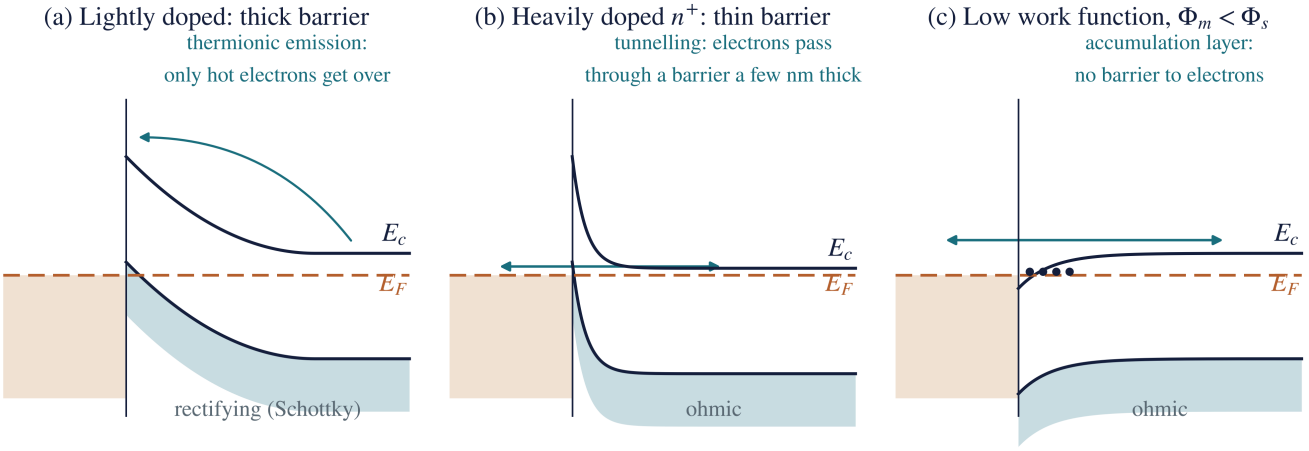


Figure 9. (a) A lightly doped contact has a thick barrier that electrons can only cross by thermionic emission: rectifying. (b) A heavily doped n^+ contact has a barrier so thin that electrons tunnel through it: ohmic. (c) A low work-function metal ($\Phi_m < \Phi_s$) gives downward bending and an accumulation layer with no barrier: ohmic.

	Thermionic emission	Tunnelling (field emission)
How the electron crosses	over the top of the barrier	through the barrier
Needs	enough thermal energy	a thin barrier (a few nm)
Doping under the contact	light to moderate ($\le 10^{23}\ \mathrm{m}^{-3}$)	heavy ($\ge 10^{25}\ \mathrm{m}^{-3}$)
Temperature dependence	strong, $\propto T^2 \exp(-q\Phi_{Bn}/kT)$	weak
Gives	Schottky (rectifying) contact	ohmic contact

Applications of ohmic contacts: source, drain and gate connections of MOSFETs; emitter, base and collector contacts of BJTs; front and back contacts of solar cells (a poor contact adds series resistance and lowers the fill factor and output power); contacts to LEDs and lasers; every connection in an integrated circuit and every test structure.

Stability of contacts. Contacts must survive hot processing and years of use. Problems include aluminium "spiking" into shallow junctions (prevented with 1 % Si in the Al or a TiN barrier), electromigration of the metal at high current density, and reactions or native oxide at the interface that change Φ_{Bn} and ρ_c.

8. Schottky contact versus ohmic contact

Why both contacts are needed. A practical device usually contains both. A Schottky diode has its rectifying contact on one side and an ohmic contact on the other (Figure 1): the Schottky contact gives the rectification, while the ohmic contact lets the current enter or leave without an extra barrier or voltage drop. In a solar cell every contact must be ohmic; an accidental rectifying contact adds resistance, lowers the fill factor and reduces the output power.

Feature	Schottky contact	Ohmic contact
Electrical behaviour	rectifying	non-rectifying
I–V curve	exponential, asymmetric	straight line through the origin
Work-function condition (n-type)	$\Phi_m > \Phi_s$	$\Phi_m < \Phi_s$, or any metal on n^+
Doping under the contact	moderate (10^{21}–10^{23} m^{-3})	heavy ($> 10^{25}$ m^{-3})
Band bending (n-type)	upward: depletion layer	downward (accumulation) or a very thin barrier
Barrier to majority carriers	yes: $q\Phi_{Bn}$ and qV_{bi}	none, or thin enough to tunnel through
Transport	thermionic emission	free flow or tunnelling
Resistance	depends on bias and polarity	small and constant
Advantages	fast switching; low forward voltage	low resistance; efficient current collection
Limitations	higher leakage; lower breakdown voltage	needs heavy doping and careful processing
Used for	rectifiers, detectors, clamps, MESFET gates	connecting every device to the circuit

9. Worked examples

All examples are for silicon at 300 K: $\chi = 4.05$ eV, $E_g = 1.12$ eV, $kT = 0.02585$ eV, $N_c = 2.8 \times 10^{25}$ m^{-3}, $\varepsilon_s = 1.04 \times 10^{-10}$ F m^{-1}, $q = 1.6 \times 10^{-19}$ C.

Example 1: work function of doped silicon

Question. Find the work function of n-type silicon with $N_D = 1 \times 10^{22}$ m^{-3}.

Given: $N_D = 1 \times 10^{22}$ m^{-3}, $N_c = 2.8 \times 10^{25}$ m^{-3}, $\chi = 4.05$ eV. **Find:** Φ_s.

Formula: $E_c - E_F = kT \ln(N_c/N_D)$, then $q\Phi_s = q\chi + (E_c - E_F)$.

Solution.

$$E_c - E_F = (0.02585)\ln\left(\frac{2.8\times10^{25}}{1\times10^{22}}\right) = (0.02585)(7.94) = 0.205\ \text{eV}$$

$$\Phi_s = 4.05 + 0.205 = 4.26\ \text{eV}$$

Answer: $\Phi_s = 4.26$ eV. **What this means:** any metal with $\Phi_m > 4.26$ eV (gold, nickel, platinum) tends to form a Schottky barrier on this silicon.

Example 2: an ideal gold contact

Question. Gold ($\Phi_m = 5.10$ eV) is deposited on the silicon of Example 1. Using the Schottky–Mott model, find the barrier height, built-in potential, depletion width and peak field at zero bias.

Given: $\Phi_m = 5.10$ eV, $\chi = 4.05$ eV, $\Phi_s = 4.26$ eV, $N_D = 1\times10^{22}\ \text{m}^{-3}$, $V = 0$. **Find:** Φ_{Bn}, V_{bi}, W, $\mathscr{E}_{\max}$.

Formulas: $q\Phi_{Bn} = q(\Phi_m - \chi)$; $qV_{bi} = q(\Phi_m - \Phi_s)$; $W = \sqrt{2\varepsilon_s V_{bi}/qN_D}$; $\mathscr{E}_{\max} = qN_D W/\varepsilon_s$.

Solution.

$$\Phi_{Bn} = 5.10 - 4.05 = 1.05\ \text{eV}, \qquad V_{bi} = 5.10 - 4.26 = 0.84\ \text{V}$$

$$W = \sqrt{\frac{2\,(1.04\times10^{-10})\,(0.84)}{(1.6\times10^{-19})\,(1\times10^{22})}} = 3.3\times10^{-7}\ \text{m}$$

$$\mathscr{E}_{\max} = \frac{(1.6\times10^{-19})(1\times10^{22})(3.3\times10^{-7})}{1.04\times10^{-10}} = 5.1\times10^{6}\ \text{V m}^{-1}$$

Answer: $\Phi_{Bn} = 1.05$ eV, $V_{bi} = 0.84$ V, $W = 3.3\times10^{-7}$ m (0.33 µm), $\mathscr{E}_{\max} = 5.1\times10^{6}\ \text{V m}^{-1}$. **What this means:** the barrier is 0.33 µm thick, far too thick to tunnel through, so the contact rectifies. In practice Fermi-level pinning gives a measured $\Phi_{Bn} \approx 0.80$ eV for gold on n-Si.

Example 3: the depletion width under bias

Question. For the gold contact of Example 2, find the depletion width at a forward bias of 0.30 V and at a reverse bias of 5 V.

Given: $V_{bi} = 0.84$ V, $N_D = 1\times10^{22}\ \text{m}^{-3}$, $\varepsilon_s = 1.04\times10^{-10}\ \text{F m}^{-1}$; $V = +0.30$ V and $V = -5$ V. **Find:** W in each case.

Formula: $W = \sqrt{2\varepsilon_s(V_{bi} - V)/qN_D}$.

Solution.

$$W_{\text{fwd}} = \sqrt{\frac{2\,(1.04\times10^{-10})\,(0.84 - 0.30)}{(1.6\times10^{-19})\,(1\times10^{22})}} = 2.65\times10^{-7}\ \text{m}$$

$$W_{\text{rev}} = \sqrt{\frac{2\,(1.04\times10^{-10})\,(0.84 + 5)}{(1.6\times10^{-19})\,(1\times10^{22})}} = 8.71\times10^{-7}\ \text{m}$$

Answer: $W = 0.265\ \mu$m at 0.30 V forward and $0.871\ \mu$m at 5 V reverse (against $0.330\ \mu$m at zero bias). **What this means:** forward bias narrows the depletion region and lowers the semiconductor-side barrier; reverse bias widens it, $\sqrt{5.84/0.84} = 2.6$ times here, which also lowers the junction capacitance.

Example 4: barrier height from the work function

Question. A metal with $\Phi_m = 4.70$ eV contacts n-type silicon. Find the ideal barrier height.

Formula: $q\Phi_{Bn} = q(\Phi_m - \chi)$. **Solution:** $\Phi_{Bn} = 4.70 - 4.05 = 0.65$ eV.

Answer: $\Phi_{Bn} = 0.65$ eV. **What this means:** an electron in the metal needs at least 0.65 eV, about 25 times kT, to enter the semiconductor; only a tiny fraction have it, which is why the reverse current is small.

Example 5: saturation current and turn-on voltage

Question. A Schottky diode on n-type silicon has a measured barrier height of 0.80 eV. Find the saturation current density at 300 K, and the forward voltage needed for $J = 1 \times 10^4$ A m^{-2} (1 A per square centimetre).

Given: $\Phi_{Bn} = 0.80$ V, $A^{**} = 1.1 \times 10^6$ A m^{-2} K^{-2}, $T = 300$ K, $kT/q = 0.0259$ V, $\eta = 1$. **Find:** J_s and V.

Formulas: $J_s = A^{**} T^2 \exp(-q\Phi_{Bn}/kT)$; for $J \gg J_s$, $V = (kT/q)\ln(J/J_s)$.

Solution.

$$J_s = (1.1 \times 10^6)(300)^2 \exp\left(-\frac{0.80}{0.0259}\right) = (9.9 \times 10^{10})(3.85 \times 10^{-14}) = 3.8 \times 10^{-3} \text{ A m}^{-2}$$

$$V = (0.0259)\ln\left(\frac{1 \times 10^4}{3.8 \times 10^{-3}}\right) = (0.0259)(14.8) = 0.38 \text{ V}$$

Answer: $J_s = 3.8 \times 10^{-3}$ A m^{-2}; the diode carries 1×10^4 A m^{-2} at about 0.38 V. **What this means:** a silicon p–n diode needs about 0.7 V for the same current density, because its saturation current density is many orders of magnitude smaller. This is why Schottky diodes are used as low-loss rectifiers.

Example 6: forward current of a Schottky diode

Question. An ideal Schottky diode ($\eta = 1$) has a saturation current of 2 μA. Find the current at a forward voltage of 0.20 V at 300 K.

Given: $I_s = 2 \times 10^{-6}$ A, $V = 0.20$ V, $kT/q = 0.02585$ V. **Find:** I.

Formula: $I = I_s[\exp(qV/\eta kT) - 1]$.

Solution. The exponent is $qV/kT = 0.20/0.02585 = 7.737$, so $\exp(7.737) = 2291$:

$$I = (2 \times 10^{-6})[\exp\left(\frac{0.20}{0.02585}\right) - 1] = (2 \times 10^{-6})(2290) = 4.58 \times 10^{-3} \text{ A}$$

Answer: $I = 4.58$ mA. **What this means:** only 0.2 V raises the current more than two-thousand-fold above I_s; every further 0.1 V multiplies it by $\exp(0.1/0.02585) \approx 48$.

Example 7: why heavy doping gives an ohmic contact

Question. The voltage across the depletion region under a contact on n-Si is 0.60 V. Find W for $N_D = 1 \times 10^{22}$ and 1×10^{25} m^{-3}.

Formula: $W = \sqrt{2\varepsilon_s(V_{bi} - V)/qN_D}$.

Solution.

$$W = \frac{\sqrt{2\,(1.04\times10^{-10})\,(0.60)}}{(1.6\times10^{-19})\,(1\times10^{22})} = 2.79\times10^{-7}\ \text{m}\ \ (279\ \text{nm})$$

$$W = \frac{\sqrt{2\,(1.04\times10^{-10})\,(0.60)}}{(1.6\times10^{-19})\,(1\times10^{25})} = 8.8\times10^{-9}\ \text{m}\ \ (8.8\ \text{nm})$$

Answer: 279 nm at $10^{22}\ \text{m}^{-3}$ and 8.8 nm at $10^{25}\ \text{m}^{-3}$ (a thousand times the doping, $\sqrt{1000} = 31.6$ times thinner). **What this means:** electrons cannot tunnel through 279 nm, so the lightly doped contact rectifies; a barrier under about 10 nm thick passes a large tunnelling current, so the heavily doped contact is ohmic, even though the barrier height is unchanged.

Example 8: a contact on p-type silicon

Question. Aluminium ($\Phi_m = 4.28$ eV) is deposited on p-type silicon. Find the ideal barrier height for holes.

Formula: $q\Phi_{Bp} = E_g - q(\Phi_m - \chi)$. **Solution:** $\Phi_{Bp} = 1.12 - (4.28 - 4.05) = 0.89$ eV.

Answer: $\Phi_{Bp} = 0.89$ eV: Al on p-Si is ideally a Schottky contact, while Al on n-Si ($\Phi_{Bn} = 0.23$ eV, ideally) would be nearly ohmic. **What this means:** the same metal can be rectifying on one type of silicon and ohmic on the other.

10. Examination question solved

Question 49. *Differentiate between a Schottky barrier diode and an ohmic contact. Explain their energy-band characteristics.*

What the examiner wants: the definition of each; band diagrams before contact, at equilibrium and under bias for the Schottky diode; the barrier formulas; the two practical ways of making an ohmic contact; the I–V behaviour; and a comparison table.

Model answer.

Both are metal–semiconductor contacts. A **Schottky barrier diode** is a contact with a significant barrier to majority carriers, so it **rectifies**. An **ohmic contact** has no barrier that limits the current, so it conducts equally in both directions with a linear I–V characteristic, $V = IR_c$. Both are explained with band diagrams using one rule: at equilibrium the Fermi level is flat. The cases below are for a metal on n-type silicon.

(a) Before contact. Energies are measured from the common vacuum level. The semiconductor work function depends on doping, $q\Phi_s = q\chi + (E_c - E_F)$. If $\Phi_m > \Phi_s$, the semiconductor's Fermi level is higher, and electrons will flow into the metal on contact (Figure 3).

(b) Schottky barrier ($\Phi_m > \Phi_s$, moderate doping). On contact, electrons flow from the semiconductor into the metal, leaving ionised donors behind. A depletion region of width W forms, the field bends the bands **upward** towards the interface, and the flow stops when the Fermi levels align (Figure 4). Two barriers result:

$$q\Phi_{Bn} = q(\Phi_m - \chi), \qquad qV_{bi} = q(\Phi_m - \Phi_s), \qquad W = \sqrt{\frac{2\varepsilon_s\,(V_{bi} - V)}{qN_D}}$$

Under **forward bias** (metal positive) the semiconductor Fermi level rises by qV, the band bending falls to $q(V_{bi} - V)$, W narrows, and electrons flow over the barrier into the metal: the current rises exponentially. Under **reverse bias** the bending rises to $q(V_{bi} + V_R)$, W widens, and only the small flow over the fixed barrier $q\Phi_{Bn}$ remains: the saturation current (Figure 7). Current is by thermionic emission of majority carriers:

$$J = A^{**}\ T^2\ \exp\left(-\frac{q\Phi_{Bn}}{kT}\right)\left[\exp\left(\frac{qV}{\eta kT}\right) - 1\right]$$

There is no stored minority charge, so the diode switches very fast and turns on at a low voltage (0.2–0.4 V on Si), but it leaks more in reverse than a p–n diode. For gold on n-Si with $N_D = 10^{22}\ \text{m}^{-3}$: $\Phi_{Bn} = 1.05$ eV ideally (about 0.8 eV measured, because of Fermi-level pinning), $V_{bi} = 0.84$ V and $W = 0.33\ \mu$m.

(c) Ohmic contact. There are two band-diagram routes (Figure 9):

- **Accumulation ($\Phi_m < \Phi_s$ on n-type).** Electrons flow from the metal into the semiconductor and accumulate at the surface; the bands bend **downward**, and there is no barrier to electrons in either direction.
- **Heavy doping ($N_D > 10^{25}\ \text{m}^{-3}$, the practical method).** The barrier height is unchanged, but since $W \propto 1/\sqrt{N_D}$ the barrier becomes only a few nanometres thick (8.8 nm at $10^{25}\ \text{m}^{-3}$ for 0.6 V, against 279 nm at $10^{22}\ \text{m}^{-3}$), and electrons **tunnel** through it in both directions.

A good ohmic contact has a specific contact resistance $\rho_c \approx 10^{-10}$–$10^{-12}\ \Omega\,\text{m}^2$; silicides on heavily doped silicon are standard.

(d) Comparison. The table in Section 8 summarises the differences: a Schottky contact has upward band bending, a fixed barrier $q\Phi_{Bn}$, thermionic emission and a rectifying exponential I–V curve; an ohmic contact has downward bending or a tunnel-thin barrier, free flow or tunnelling, and a linear, symmetric I–V curve.

Marking checklist: definitions of each; $q\Phi_s = q\chi + (E_c - E_F)$ and the work-function condition; the formation steps (electron flow, depletion, upward bending, Fermi levels align); $q\Phi_{Bn} = q(\Phi_m - \chi)$ and $qV_{bi} = q(\Phi_m - \Phi_s)$; band diagrams before contact, at equilibrium, forward and reverse bias with $q\Phi_{Bn}$ fixed; the thermionic-emission equation; the two ohmic routes (accumulation; heavy doping and tunnelling); a comparison table.

Common mistakes: saying an ohmic contact "has no barrier" (the practical n^+ contact has one, just a very thin one); drawing Φ_{Bn} changing with bias (only the semiconductor-side barrier changes); reversing the conditions for p-type; confusing Φ_{Bn} (from the metal Fermi level) with V_{bi} (from the semiconductor conduction band); writing the band gap as $E_v - E_c$ (it is $E_c - E_v$).

11. Summary

- At equilibrium the Fermi level is flat; electrons flow from the higher Fermi level to the lower one when two materials touch.
- $q\Phi_s = q\chi + (E_c - E_F)$. A metal on n-type with $\Phi_m > \Phi_s$ forms a Schottky barrier; $\Phi_m < \Phi_s$, or heavy doping, gives an ohmic contact.
- Schottky barrier: depletion region, upward band bending, $q\Phi_{Bn} = q(\Phi_m - \chi)$, $qV_{bi} = q(\Phi_m - \Phi_s)$, $W = \sqrt{2\varepsilon_s(V_{bi} - V)/qN_D}$. In real contacts the barrier is set largely by Fermi-level pinning.
- Under bias $q\Phi_{Bn}$ stays fixed; only the semiconductor-side barrier changes, to $q(V_{bi} - V)$ or $q(V_{bi} + V_R)$.
- Current is thermionic emission of majority carriers: $J = A^{**}T^2\ \exp(-q\Phi_{Bn}/kT)[\exp(qV/\eta kT) - 1]$. Schottky diodes are fast, turn on at a low voltage and leak more than p–n diodes.
- Ohmic contacts: accumulation (low barrier), tunnelling through a thin barrier made by heavy doping ($W \propto 1/\sqrt{N_D}$), and silicide contact engineering. An ohmic contact may still have a barrier; what matters is that it does not limit the current.

Symbols used

Symbol	Meaning	SI unit
q	electronic charge, 1.6×10^{-19}	C
k	Boltzmann constant, 1.381×10^{-23}	$\mathrm{J\,K^{-1}}$
T	absolute temperature	K
$E_{\mathrm{vac}}, E_c, E_v, E_F$	vacuum level, conduction-band edge, valence-band edge, Fermi level	eV (J)
E_g	band gap, $E_c - E_v$	eV
Φ_m, Φ_s	work function of the metal, of the semiconductor	V ($q\Phi$ in eV)
χ	electron affinity	V ($q\chi$ in eV)
Φ_{Bn}, Φ_{Bp}	Schottky barrier height for electrons, for holes	V ($q\Phi_B$ in eV)
V_{bi}	built-in potential	V
V, V_R	applied voltage (positive forward); reverse voltage	V
N_D, N_A	donor, acceptor concentration	$\mathrm{m^{-3}}$
N_c	effective density of states in the conduction band	$\mathrm{m^{-3}}$
W	depletion width	m
$\varepsilon_0, \varepsilon_s$	permittivity of free space (8.854×10^{-12}), of the semiconductor	$\mathrm{F\,m^{-1}}$
$\mathscr{E}, \mathscr{E}_{\max}$	electric field, its peak value	$\mathrm{V\,m^{-1}}$
ρ	charge density	$\mathrm{C\,m^{-3}}$
J, J_s	current density, saturation current density	$\mathrm{A\,m^{-2}}$
I, I_s	current, saturation current	A
A^{**}	effective Richardson constant	$\mathrm{A\,m^{-2}\,K^{-2}}$
η	ideality factor	—
A	contact area	$\mathrm{m^2}$
R_c, ρ_c	contact resistance; specific contact resistance	Ω; $\Omega\,\mathrm{m^2}$
$\Delta\Phi$	image-force barrier lowering	V

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